

IKRAMUL HASAN

Kanpur, Uttar Pradesh, India | +91-9596455297 | ikramul7388@gmail.com

[LinkedIn](#) | [LeetCode](#) | [GitHub](#)

Summary

Final-year Information Technology undergraduate at NIT Srinagar interested in Software Development, and Distributed Systems. Hands-on experience building scalable, reliable systems using modern technologies.

Skills

Programming Languages: C++(Primary), C, SQL, JavaScript, TypeScript

Backend & web: HTML, CSS, React, Node.js, Express.js, MongoDB, Socket.io, Git/GitHub, Postman

Distributed Systems: Queue Broker(RabbitMQ), Redis, Docker, Microservice Architecture

AI Tools & Automation: LLM Integration(Gemini, OpenAI, Groq), Playwright(Script Automation)

Core CS: Data Structures & Algorithms(DSA), Object-Oriented Programming(OOP), Database Management system(DBMS), Operating System(OS), Computer Networks(CN), Software Development Life Cycle(SDLC)

Experience

IIT Mandi — IKSMHA Centre

Dec 2025 – Jan 2026

Research & Development Intern (VR + AI, Multimodal Comprehension)

On-site

- Led design of the experimental framework and narrative architecture across multiple storytelling conditions.
- Fabricated detailed VR narrative spaces, customizing character designs and animations to specifically address and measure 4 distinct user behavior hypotheses(text, audio, VR, robot-assisted).
- Developed web components using the MERN stack to support the digital infrastructure and experimental workflow.

Flipr Generative AI National Hackathon (Github Link)

July 2025

Software Developer

- Collaborated in a 4-member team that secured **All India Rank 1** among 64+ competing teams nationwide.
- KR.AI (Autonomous Meeting Intelligence Agent): Introduced an AI-driven tool that converts unstructured meeting conversations into structured insights, action items, and summaries using GenAI and LLMs.
- Created a Chrome Extension for real-time Zoom call capture, with AssemblyAI for transcription and RAG-based contextual memory.

Tech Stack: React, Tailwind CSS, FastAPI, OpenAI, AssemblyAI, OpenRouter, Render, Vercel.

HackerRank Infinity Hacks Global Hackathon (Github Link)

August 2026

Software Developer

- Collaborated in a 5-member team that secured **Global Rank 20** among 1500+ teams and 6000+ developers, design and ship an end-to-end Graph-RAG intelligence platform from data ingest to analyst UI.
- Developed Narco-Graph Intel, a Graph-RAG anti-narcotics OSINT system that converts scattered public reporting into a Neo4j knowledge graph of people, drugs, places, and cases using GenAI and LLMs.
- Built adaptive retrieval with Voyage embeddings, constrained graph traversal, and auto web-crawl (SearXNG/Scrapy) when confidence is low, plus Cypher risk flags for multi-city actors and recurring co-accused.

Tech Stack: FastAPI, Next.js, Neo4j(Graph, Vectors), Voyage(Embeddings), Groq(LLM), SearXNG(Search), Scrapy(Crawl).

Projects

Ai Powered Autonomous Distributed Website Testing Platform (Github Link)

- Architected a 6-microservice QA platform (API Gateway, Auth, AI, Execution, Report, Notification) exposing 33 REST APIs, containerized as 10 Docker services - replacing monolithic coupling with independently deployable, independently scalable domains.
- Introduced a RabbitMQ-based distributed Playwright execution engine (3 queues + DLQ, prefetch=5), enabling up to 5× concurrent job throughput over sequential single-worker execution, with bounded retry logic (≤ 3 attempts) before dead-lettering failed jobs.
- Integrated Gemini 2.5 Flash to generate Playwright test (11 action, 7 locator, 6 assertion); added Redis caching (TTL 3600s) keyed on prompt+URL, eliminating redundant AI calls on cache hits 100% cost and latency savings on repeat runs.
- Delivered real-time execution visibility via Socket.io (live progress, logs, notifications) with a 3s polling fallback, removing the need for manual refresh when monitoring long-running Playwright jobs.
- Hardened platform security with short-lived JWTs (15-min access / 7-day refresh), Redis-backed session revocation, and gateway-level rate limiting (200 req/15min in prod) to mitigate brute-force and API abuse.

Tech Stack: Node.js, TypeScript, Express, React, MongoDB, Redis, RabbitMQ, Playwright, Docker, Socket.io

RISC-V CPU Simulator (Github Link)

- Designed a cycle-accurate RISC-V CPU simulator in C++ modeling the classic 5-stage pipeline (IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM $\rightarrow$ WB) with a 32×32-bit register file and 1,024-word data memory.
- Built a 2-pass RISC-V parser supporting 6 instructions across 4 formats (R/I/S/B): ADD, SUB, ADDI, LW, SW, BEQ — with label resolution and branch handling.
- Implemented per-cycle execution tracing logging all 5 pipeline stages per instruction, enabling step-by-step debug of arithmetic, load/store, and conditional branch behavior.
- Assembled gnuplot-based register visualization, auto-exporting non-zero register values to reg_values.dat for post-execution analysis.

Achievements

- Secured Global Rank 1119 in LeetCode Weekly Contest 506 (36.7K+ participants).
- Selected for GSSoC 2025 & 2026 - national-level open-source contribution program.
- Solved 700+ DSA problems across LeetCode(Rating: 1703), GFG, and Coding Studio; maintained 250+ Day LeetCode Streak.
- Scored 99.1 percentile in Mathematics and 97.23 overall percentile in JEE Mains.

Education

National Institute of Technology (NIT) Srinagar

Aug 2023 – Jul 2027*

B.Tech, Information Technology — CGPA: 7.5/10 | No Active or Passive Backlog

VVM Inter College, Jahanabad, Fatehpur

Aug 2018 – May 2022

Senior Secondary (82.4%) | Secondary (85.8%)